

Jesse Yu

Senior Software Engineer

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SKILL

Frontend: HTML, CSS, SASS, JavaScript, TypeScript, React.js, React Native, Vue.js, Next.js, Vite, Redux, Zustand, Context API, Tailwind CSS, Bootstrap, Material UI, Ant Design, Vuetify, D3.js, ApexCharts.js, Chart.js, Framer Motion, React Hook Form, Storybook, Styled Components

Backend: Node.js, Express.js, Python (Django, FastAPI), Java (Spring Boot), C# (.NET Core), GraphQL, RESTful APIs, Prisma ORM, JWT, OAuth 2.0, Kafka, Redis, RabbitMQ, WebSockets, Celery, gRPC, Entity Framework, Hibernate, SQLAlchemy

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, SQL Server, SQLite, Cosmos DB, Redis

AI / ML / LLMs: OpenAI (Chat API, Assistants, Embeddings), LangChain, Pinecone, 11labs, Vector Databases, Hugging Face Transformers, RAG Pipelines, Prompt Engineering, Function Calling, Semantic Search

DevOps / Infrastructure: AWS (EKS, Lambda, S3, ECS, CloudFront, Cognito, KMS, Route53, CloudWatch, SNS, SQS), GCP, Azure, Docker, Kubernetes, Jenkins, GitHub Actions, CircleCI, ArgoCD, Terraform, Pulumi, Helm, CloudFormation, NGINX, PM2, Netlify, Vercel Testing & QA: Jest, Cypress, Playwright, Enzyme, Vitest, JUnit, NUnit, MSTest, React Testing Library, Postman, Newman, Supertest, A/B Testing, Feature Flagging (LaunchDarkly)

Architecture & Patterns: Monorepo, Microservices, Serverless, Atomic Design, Event-Driven Architecture, Clean Architecture, Modular Design, Layered Architecture, Domain-Driven Design (DDD)

APIs & Integration: Stripe, SendGrid, Twilio, Shopify, Magento, WebRTC, Mapbox GL, Google Maps API, Apollo Client, React Query, Swagger/OpenAPI, REST Hooks, Socket.IO, Auth0, Clerk, Firebase Auth

Analytics & Monitoring: Datadog, Sentry, New Relic, Amplitude, Mixpanel, Google Analytics, Google Tag Manager, Prometheus, Grafana, Jaeger, LogRocket, PagerDuty, OpenTelemetry

Design & Prototyping: Figma, Sketch, Adobe XD, Zeplin, Canva

CI/CD & Source Control: Git, GitHub, Bitbucket, NX, Lint-Staged, ESLint, Prettier, Husky, Turborepo

Project Management & Collaboration: Jira, Confluence, Notion, Slack

Professional Experience

Stripe — Senior Software Engineer

Mar 2020 - Present

- Contributed across several high-impact internal and external-facing projects, including:
 - The **internal AI Chat application ("LLM Explorer")** that enables employees to interact with documentation, logs, support tools, dashboards, and customer data using natural language.

- The development of **Stripe Link**, to enable seamless one-click global payments across over 185 countries, improving checkout conversion and usability
- Work on **Stripe Treasury**, allowing platform partners to offer embedded bank accounts, card services, and cash management via APIs — powering embedded finance experiences for platforms like Shopify.
- Work on the **Payments Foundation Model**, an AI-driven model trained on billions of transactions to improve fraud detection (reducing card testing by over 60% for large businesses), authorization rates, and checkout personalization.
- Engineered backend microservices using **TypeScript (Node.js)**, **Go**, and **Java**, deployed to **AWS EKS (Kubernetes)**, with infrastructure managed via **Terraform**, **Pulumi**, and **Helm**.
- Built interactive front-ends using **React**, **Next.js**, **Tailwind CSS**, **TypeScript**, **Zustand**, **TanStack Query**, **GraphQL**, and **Apollo Client**, focusing on accessibility, performance, and secure data access.
- Devised AI-based features using **OpenAI Assistant API**, **LangChain**, **Pinecone**, and **vector embeddings** to implement semantic search, language understanding, summarization, and function-call workflows in internal tools.
- Designed relational and NoSQL data models using **Prisma**, **PostgreSQL**, **MySQL**, and **Redis**. Handled complex migrations and schema evolution via **Liquibase** and **Ghostferry**.
- Built event-driven workflows using **Kafka**, **AWS SQS**, **SNS**, and **Step Functions**, powering asynchronous backend tasks, notifications, and transaction orchestration.
- Established CI/CD pipelines with **GitHub Actions**, **CircleCI**, and **ArgoCD**; launched feature flags and experiments with **LaunchDarkly** for safe rollout and staged deployments.
- Instrumented production services using **OpenTelemetry**, **Prometheus**, **Grafana**, and **Jaeger**, setting up health dashboards and performance alerts that significantly reduced mean time to detection and repair.
- Authored comprehensive developer documentation (OpenAPI, internal Readme.io), led peer code reviews, mentored junior engineers, and helped shape architecture discussions to support scaling across teams.

Third Eye Health — Full Stack Developer

Sep 2015 - Oct 2019

- Joined as a junior developer and grew into a core contributor for Third Eye Health's **post-acute telehealth platform**, used by nurses and clinicians to provide 24/7 virtual care for patients in skilled nursing facilities. The platform enabled **secure video consultations**, **real-time messaging**, **patient intake workflows**, **e-prescriptions**, and **care escalation tools**, helping reduce hospital readmissions and improving access to overnight care.
- Built and maintained full-stack features using **C# / .NET**, **.NET Core**, along with **Python (Django)** and **Node.js** for backend APIs. Developed responsive, accessible frontend components with **AngularJS** (initially) and later **React**, **Redux**, and **JavaScript**, introducing **TypeScript** adoption in later phases.

- Integrated **WebRTC** for live, HIPAA-compliant video calls and **Twilio** for SMS and call notifications. Implemented custom **STUN/TURN** server logic to ensure stable, low-latency connectivity between clinicians and facility staff.
- Designed and optimized relational data models in **PostgreSQL** and **SQL Server**; implemented caching and real-time updates with **Redis** and **WebSockets**; and built asynchronous background jobs using **Celery**.
- Participated in migrating parts of the monolithic application into microservices and strengthened API security with **OAuth 2.0**, **JWT**, and **AWS Cognito**.
- Contributed to infrastructure automation and deployments using **Docker**, **Jenkins**, **AWS ECS**, **S3**, and **CloudFront**, introducing **CloudFormation** for infrastructure-as-code in later stages.
- Integrated third-party services including **Stripe** for co-pays, **PointClickCare** for EHR synchronization, and **SendGrid** for transactional email. Developed event-driven workflows using **AWS Lambda**, **SNS**, and **SQS** for alerts and reporting.
- Enhanced observability with **Datadog**, **Sentry**, and **AWS CloudWatch**; implemented structured logging to speed up production debugging and reduce mean time to resolution.
- Contributed to **HIPAA compliance** efforts, including data encryption with **AWS KMS**, role-based access control, and security reviews for all new features.

Education

University of Illinois Chicago

2011 - 2015

Bachelor's Degree of Computer Science